Aircraft State Estimation

CEC 300: Computing in Aviation and Aerospace

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Outline

- State variables
- State Transformations
- State Estimation

Module Introduction

- Within this module, you shall learn how unmanned aircraft and spacecraft are controlled utilizing a control system.
  - Requirements must be derived from the mission profile to determine the control modes that must be supported by the control system.
  - Onboard sensors combined with sensor fusion provides robust estimates of the vehicle's state.
    - Kalman Filters will be discussed as one such technique.
  - Control systems will be reviewed for:
    - Aircraft lateral and vertical control for Uncrewed Aircraft Systems.
    - Attitude control of a spacecraft through an Attitude Detection and Control System (ADCS).

Example: UAS States Parameters and Control Channels

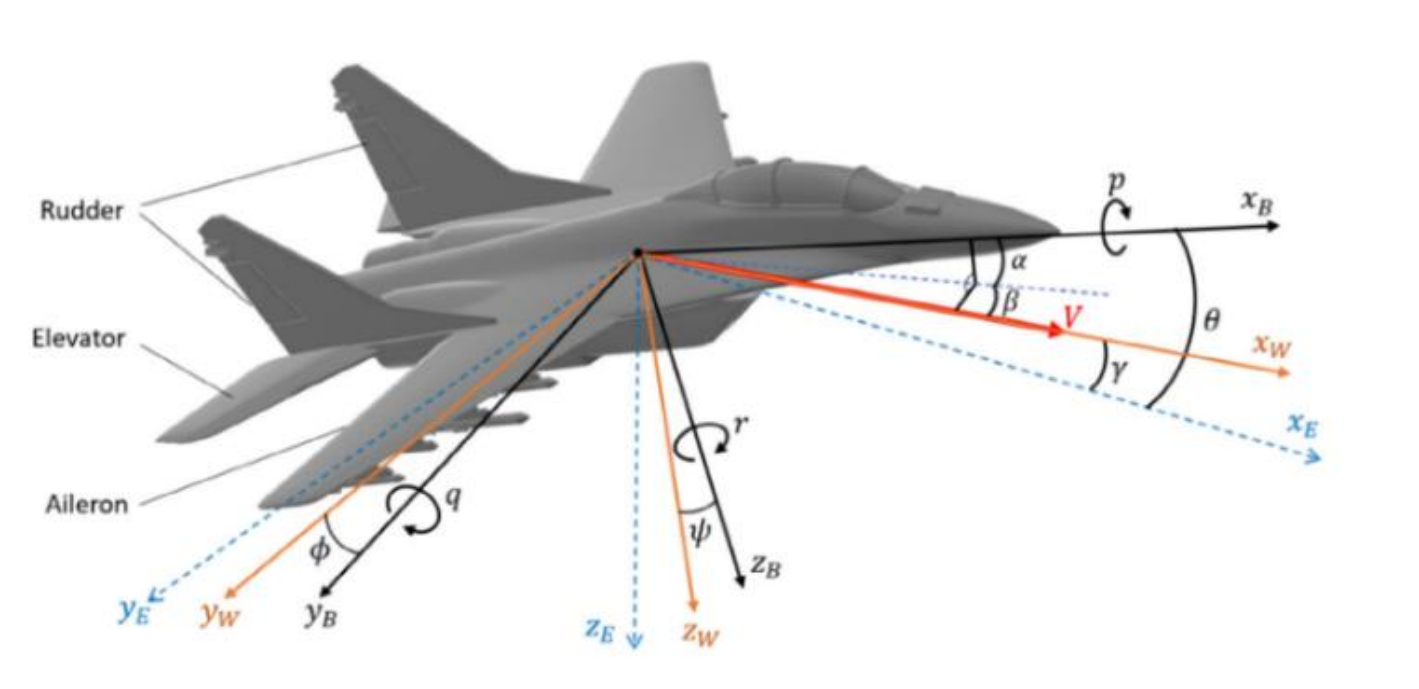


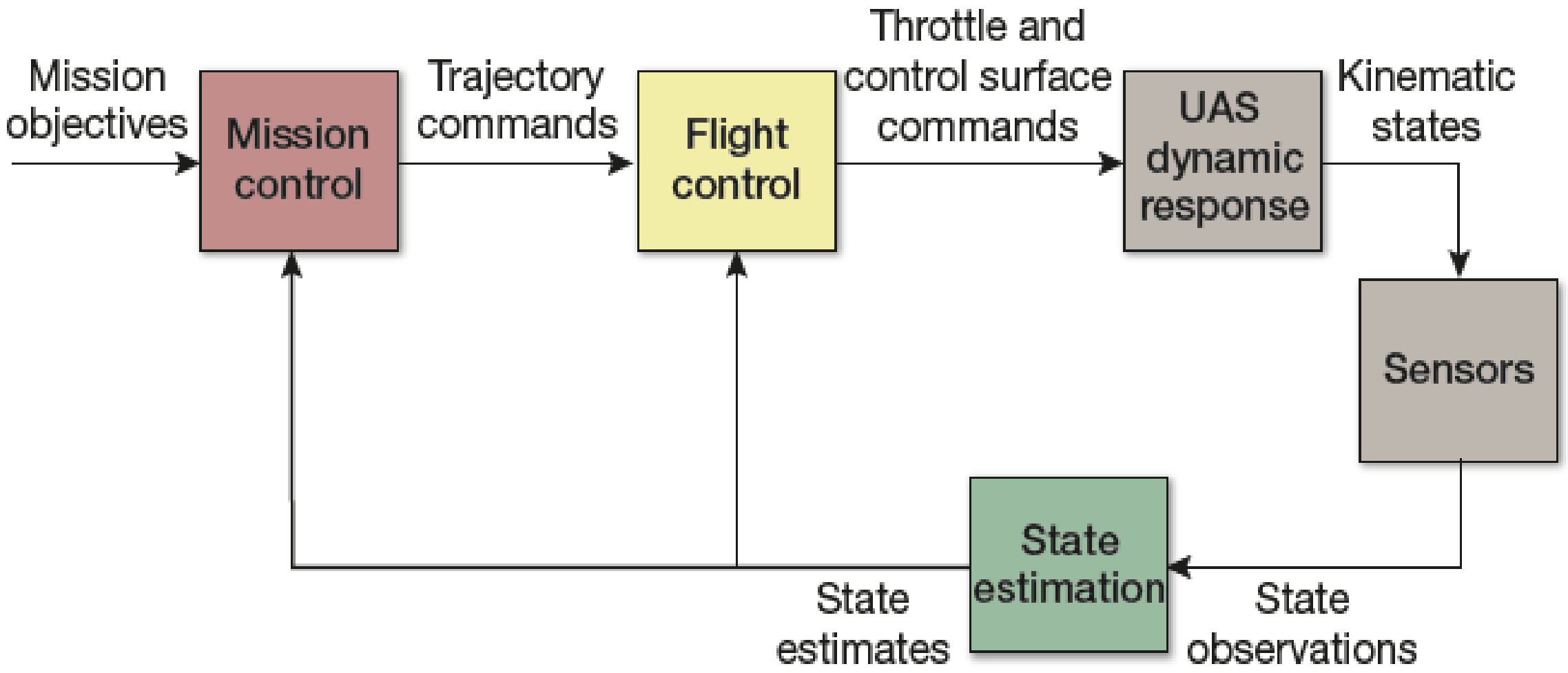
Table 10.1 UA State Parameters

	State variable	Notes
Position in Earth coordinates	x_E	Relative to a fixed longitude, μ
	y_E	Relative to a fixed latitude, λ
	z_E	Relative to the Earth's surface
Airspeed	V	Total airspeed
	u	Airspeed along x axis (body)
	v	Airspeed along y axis (body)
	w	Airspeed along z axis (body)
Aerodynamic angles	α, Angle of attack	
	β, Angle of sideslip	
Euler angles	Ψ, Heading angle	Relative to North, positive in West direction (counterclockwise)
	θ, Pitch angle	Relative to horizontal, positive up
	φ, Roll angle	
Accelerations in body axes	a_x	
	a_y	
	a_z	
Angular rates in body axes	p	Roll rate
	q	Pitch rate
	r	Yaw rate
Geographic position	λ, Latitude	Can be related to y_E
	μ, Longitude	Can be related to x_E

Four common control channels:

- throttle (T channel),
- elevator (δ_E channel),
- aileron (δ_A channel), and
- rudder (δ_E channel)

Example UAS Flight Control Software Block Diagram



Lesson Introduction

- As discussed earlier in the course, there are a variety of sensors whose inputs provide a partial picture at the aircraft's state.
  - **Position** – GPS, Accelerometer, Air data system
  - **Velocity** – GPS, Air Data System, accelerometer
  - **Acceleration** – Accelerometer
  - **Roll** – Accelerometer, thermopile, multiple GPS receivers
  - **Yaw** – magnetometer and/or gyroscope
  - **Pitch** – Altimeter and accelerometer
  - **Encoders** – measure actual position of rudder, aileron, flaps, etc.

- For a control system to work accurately, an approach is needed to combine the data from the various data sources to improve the estimated state of the aircraft **(i.e. sensor fusion)**

- Motivation for multi-sensor fusion:
  - Redundancy
  - State estimation error reduction
  - Different update rates

STATE PARAMETERS AND ESTIMATION

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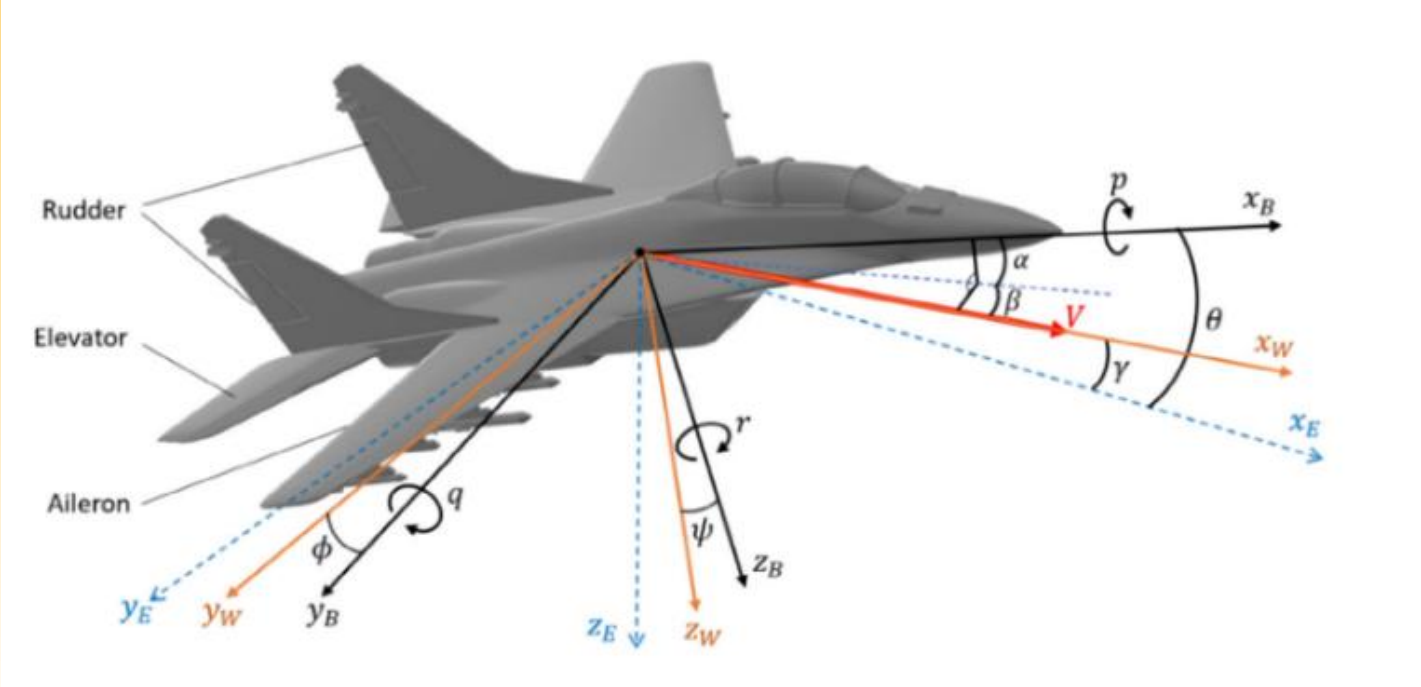
Body-Frames

- Body-Frame:
  - x_b - vector from the aircraft's center of gravity to the nose.
  - y_b - vector perpendicular to x_b along the plane made by the aircraft's horizontal body toward wings.
  - z_b - vector pointing down from the aircraft's body perpendicular to the x_b and y_b plane



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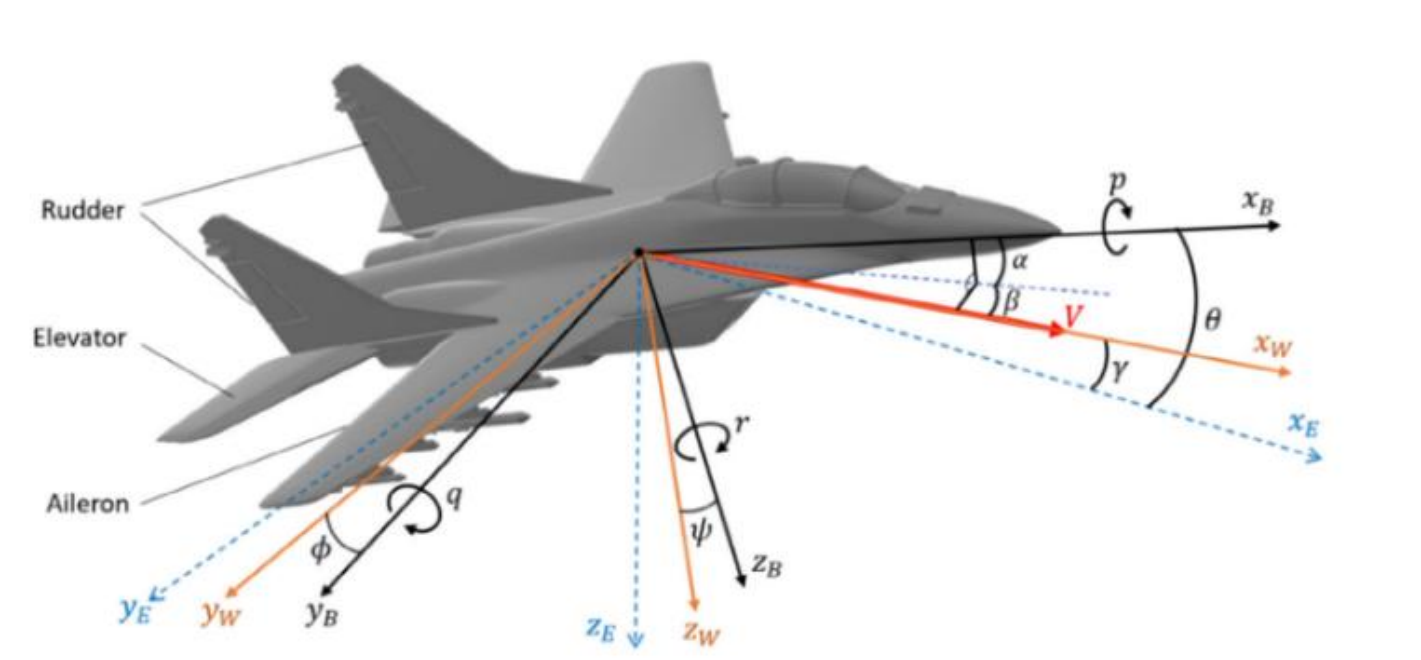
North-East-Down (NED) Frame

Origin at the a local point on the Earth Surface.

- x_E - points to True North
- y_E - perpendicular to the x_e and z_e plane pointing East
- z_E - vector pointing down toward the center of the earth.
- x_e, y_e- plane is tangent to earth surface.

- Moves with the body plane.



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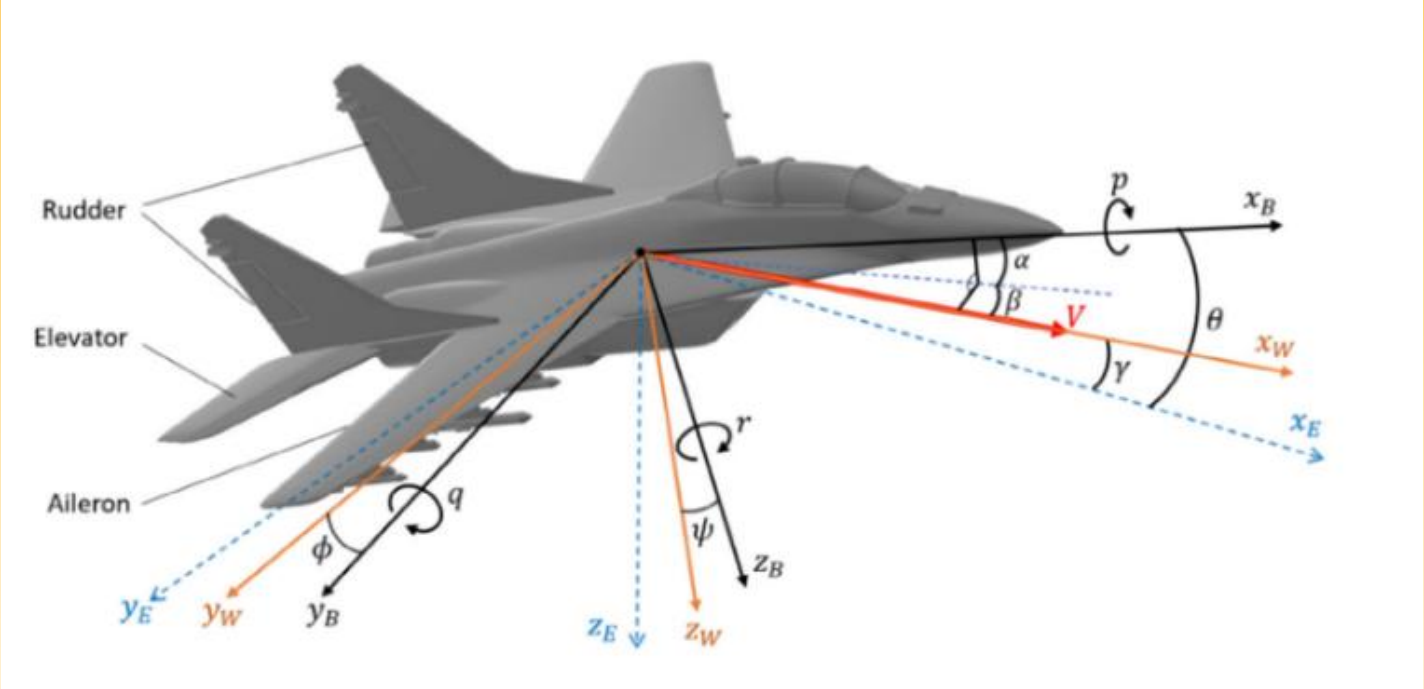
Wind Reference Frame

- x_w - points to wind vector.
- y_w - perpendicular to the x_w and z_e plane pointing East
- z_w - points downward perpendicular to the x_w axis along the aircraft's plane of symmetry



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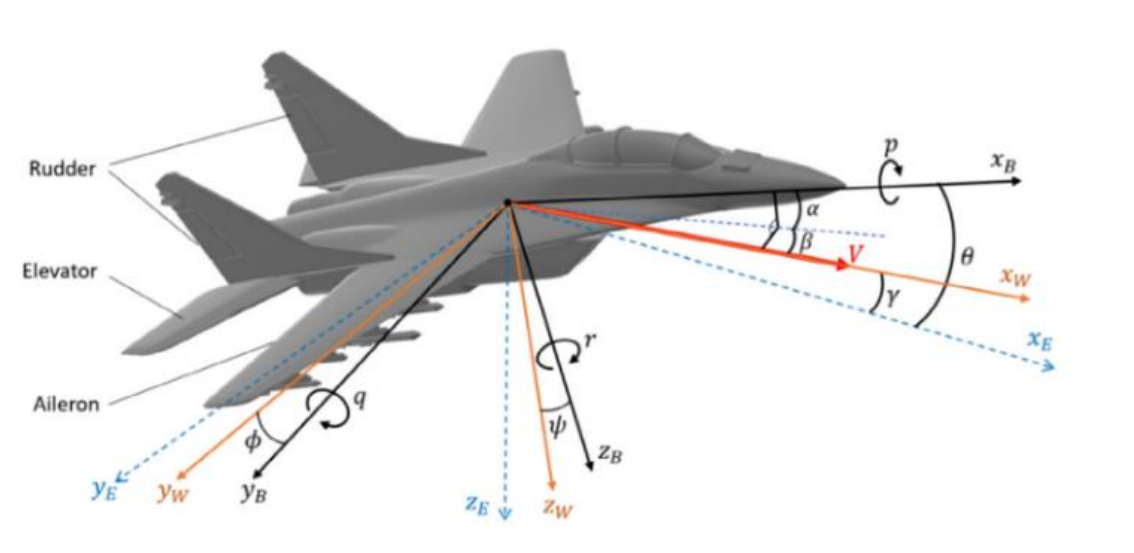
Flight Control Channels



Surface	Function(s)
Ailerons	Control roll about the longitudinal axis.
Elevator	Control pitch about the lateral axis.
Rudder	Controls yaw about the vertical axis.
Flaps	Increase lift and induced drag when deployed. Permits higher flight speeds (retracted) and lower landing speeds (extended).
Spoiler	Reduces lift and increases drag. Controls descent. Can also be used for roll control with reduced adverse yaw.
Trim	Aerodynamically assist movement and position of flight control system allowing pilot to "tune" a particular control setting allowing a more hands-off operation under R/C control. • Trim can be set within an R/C controller to define control surface position when hands-off. • Trim mechanisms can be installed on control surfaces to assist pilot in manually dialing in trim for example, trim tabs on elevators allow the pilot to adjust the hands-off pitch of the aircraft.

Four common control channels:

- throttle (T channel),
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Coordinate system and state parameters

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	μ, Longitude	Can be related to x_E

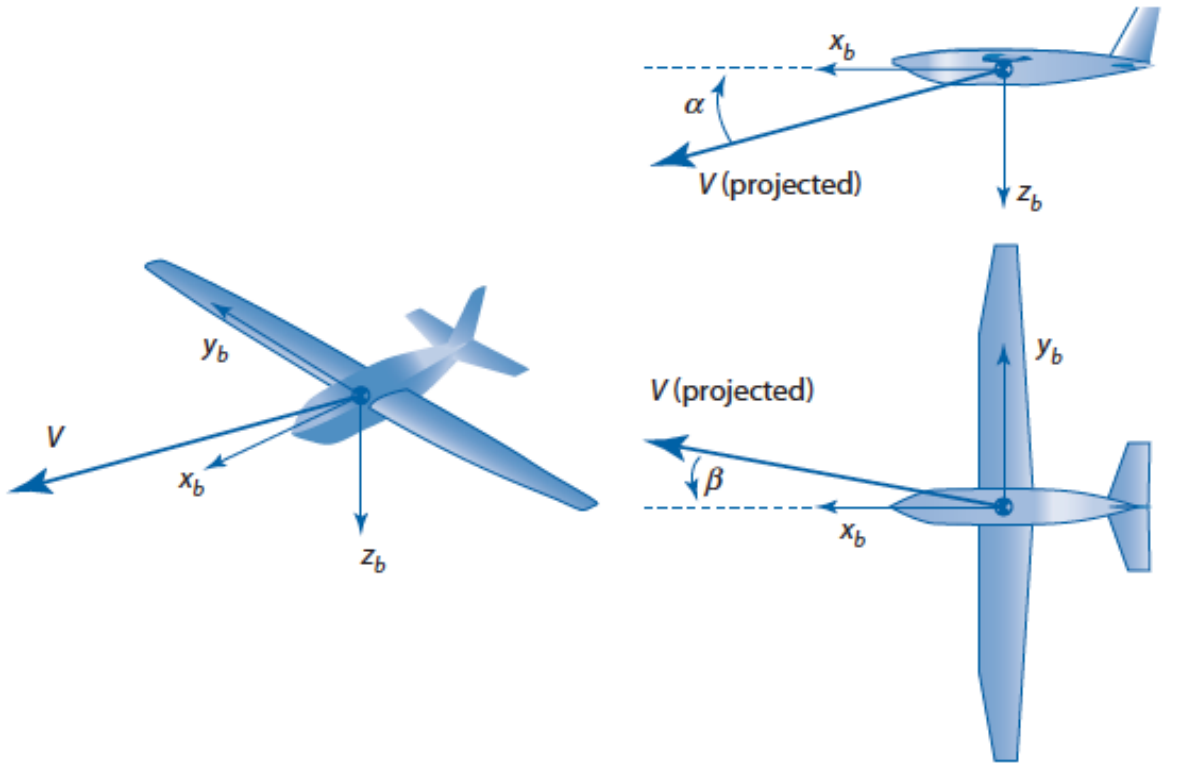


Fig. 4.3 Body-axis coordinate system.

State Calculations

Velocity in the Earth Coordinate System

$$\dot{\boldsymbol{x}}_E = \boldsymbol{R}_b^E \boldsymbol{V} + \boldsymbol{V}_w \tag{10.2}$$

Where, R_b^E is transformation from body axis to earth coordinate system, V is velocity vector in aircraft body coordinates, and V_w is the wind velocity vector

Transformation,

$$\boldsymbol{R}_b^E = \begin{bmatrix} c\theta \cdot c\psi & (s\phi \cdot s\theta \cdot c\psi - c\phi \cdot s\psi) & (c\phi \cdot s\theta \cdot c\psi + s\phi \cdot s\psi) \\ c\theta \cdot s\psi & (s\phi \cdot s\theta \cdot s\psi + c\phi \cdot s\psi) & (c\phi \cdot s\theta \cdot c\psi - s\phi \cdot c\psi) \\ -s\theta & s\phi \cdot c\theta & c\phi \cdot c\theta \end{bmatrix} \tag{10.3}$$

Notation:
s is sine
c is cosine

Velocity equation expanded out:

$$\begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{bmatrix} = \begin{bmatrix} c\theta \cdot c\psi & (s\phi \cdot s\theta \cdot c\psi - c\phi \cdot s\psi) & (c\phi \cdot s\theta \cdot c\psi + s\phi \cdot s\psi) \\ c\theta \cdot s\psi & (s\phi \cdot s\theta \cdot s\psi + c\phi \cdot s\psi) & (c\phi \cdot s\theta \cdot c\psi - s\phi \cdot c\psi) \\ -s\theta & s\phi \cdot c\theta & c\phi \cdot c\theta \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} + \begin{bmatrix} V_{w,x} \\ V_{w,y} \\ V_{w,z} \end{bmatrix} \tag{10.4}$$

$\dot{x}_E$ is north velocity
$\dot{y}_E$ is east velocity
$\dot{z}_E$ is altitude change rate

State Calculations (2)

- Airspeed along x, y, and z axis (u, v, and w) can be derived using an air data system's measurements of sideslip (β), angle of attack (α), and observed velocity (V).

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} V \cdot c\alpha \cdot c\beta \\ V \cdot s\beta \\ V \cdot s\alpha \cdot c\beta \end{bmatrix}$$

- Horizontal wind component (if wind is acting on horizontal plane) can be calculated using heading angle and total observed wind.

$$\boldsymbol{V}w = \begin{bmatrix} V_w \cdot s\psi_w \\ V_w \cdot c\psi_w \\ 0 \end{bmatrix}$$

Vertical element may be needed for z-component, which can be added to this matrix if known

State Calculations (3)

- Euler angular rates (change in roll, pitch, and heading) can be derived by the angular rates of the vehicle's body (roll, pitch, and, yaw rates)

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & s\phi \cdot t\theta & c\phi \cdot t\theta \\ 0 & c\phi & -s\phi \\ 0 & s\phi/c\theta & c\phi/c\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

- Or, if the Euler angular rates are known, the body axes angular rates can be calculated:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -s\theta \\ 0 & c\phi & c\theta \cdot s\phi \\ 0 & -s\phi & c\theta \cdot c\phi \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

State Calculations (4)

- Rate of change of latitude and longitude can be calculated using the east velocity component or the north velocity component, respectively.
  - Current latitude is also required to estimate the longitudinal rate

$$\dot{\lambda} = \frac{\dot{y}_E}{R_{\text{Earth}}}$$

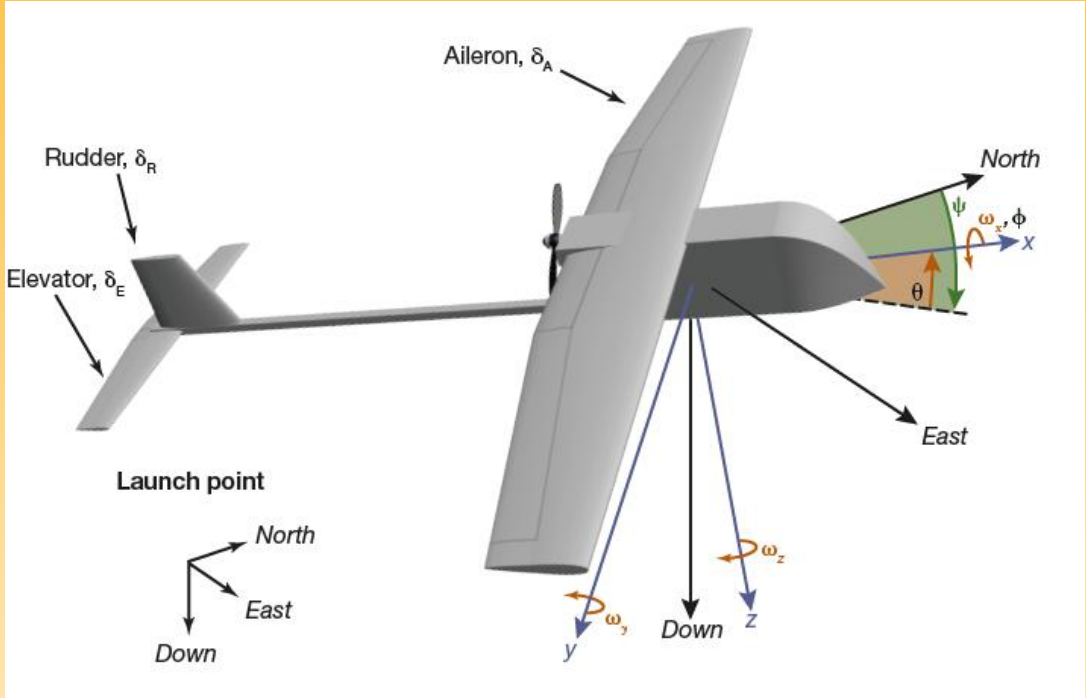
Rate of latitude change

$$\dot{\mu} = \frac{\dot{x}_E}{R_{\text{Earth}} \cdot \cos(\lambda)}$$

Rate of longitude change

$R_{Earth} = 6.378\ 100 \text{ x} 10^6$meters

Flight Control Channels



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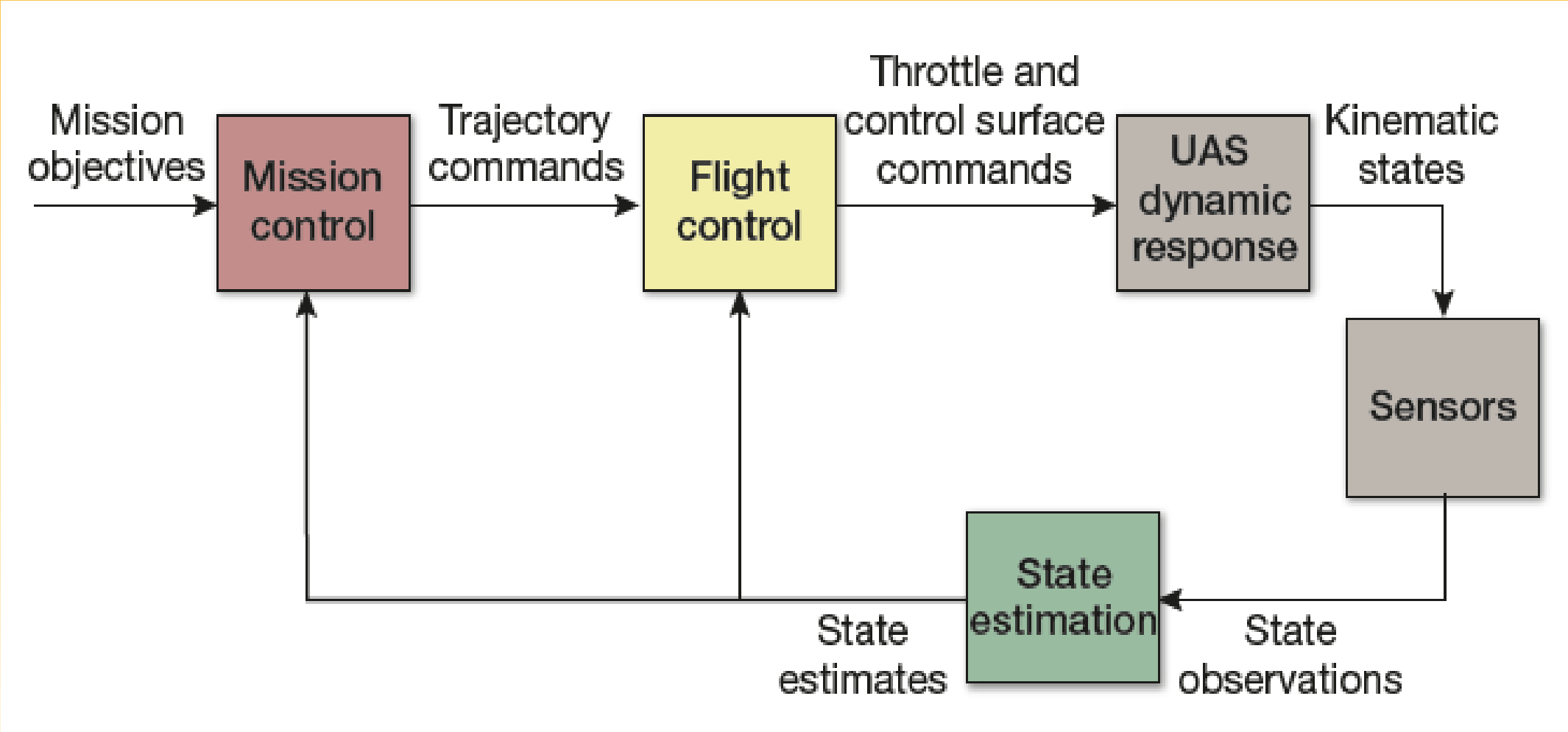
Bigger Picture - Flight Control Software – Block Diagram



STATE MEASUREMENT AND ESTIMATION

Overview

Vehicle state vector updates can come from a variety of sources:

- Accelerometers
  - Measure linear accelerations in the x, y, and z body axes (ax, ay, az)
  - Integrated with time to yield velocity (u, v, w)
  - Velocities and accelerations can be integrated with time to give position (xE, yE, zE)
- Gyroscopes
  - Measure the Euler angles (θ, φ, ϕ)
  - Angular rates from the derivative of Euler angles (p, q, r)
  - Angular acceleration from the second derivative of the Euler angles $(\dot{p}, \dot{q}, \dot{r})$
- GPS
  - Latitude and Longitude (λ, μ)
  - Position in Earth Coordinates (X_E, Y_E, Z_E)
  - Velocity (via first derivative of position over time) (u, v, w)
- Level Sensor
  - Roll (ϕ)
- Magnetic Compass
  - Euler Angles (θ, φ, ϕ)
- Air Data System
  - True airspeed (V_T)
- INS – fuses multiple sensors together

Example of Sensor Fusion

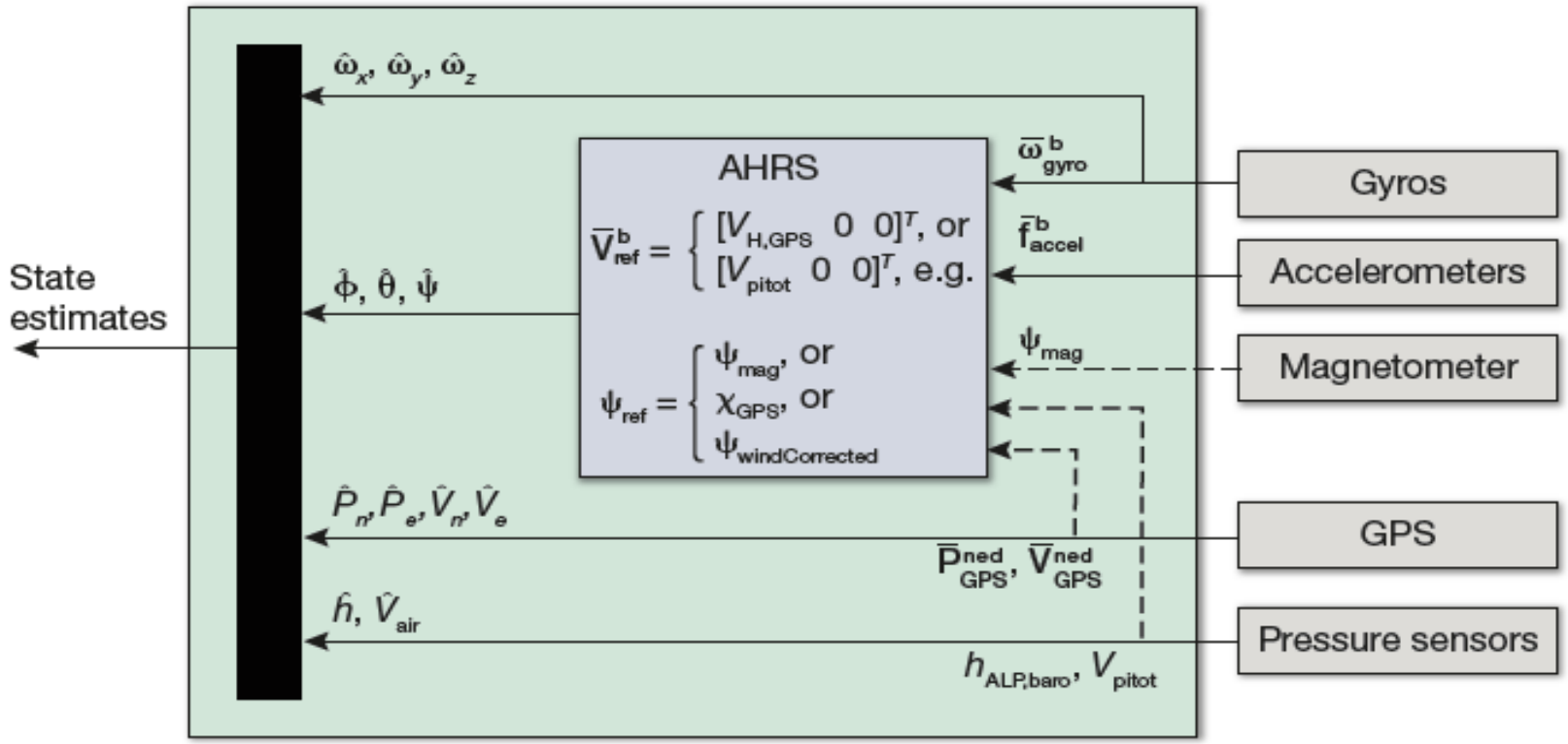


Figure 8. State estimation: GPS with an IMU-based AHRS. Position and velocity estimates are acquired from GPS and pressure sensors. The orientation is acquired by fusing gyro and accelerometer measurements with yaw and velocity references, generally via an EKF or a feedback filter.

AHRS / INS (example of state estimation)

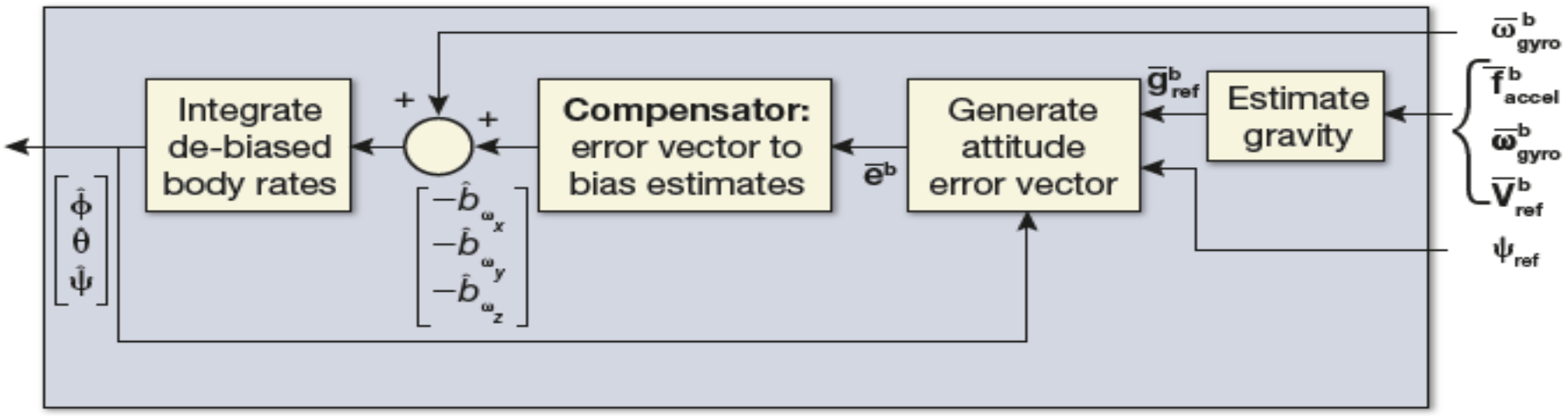
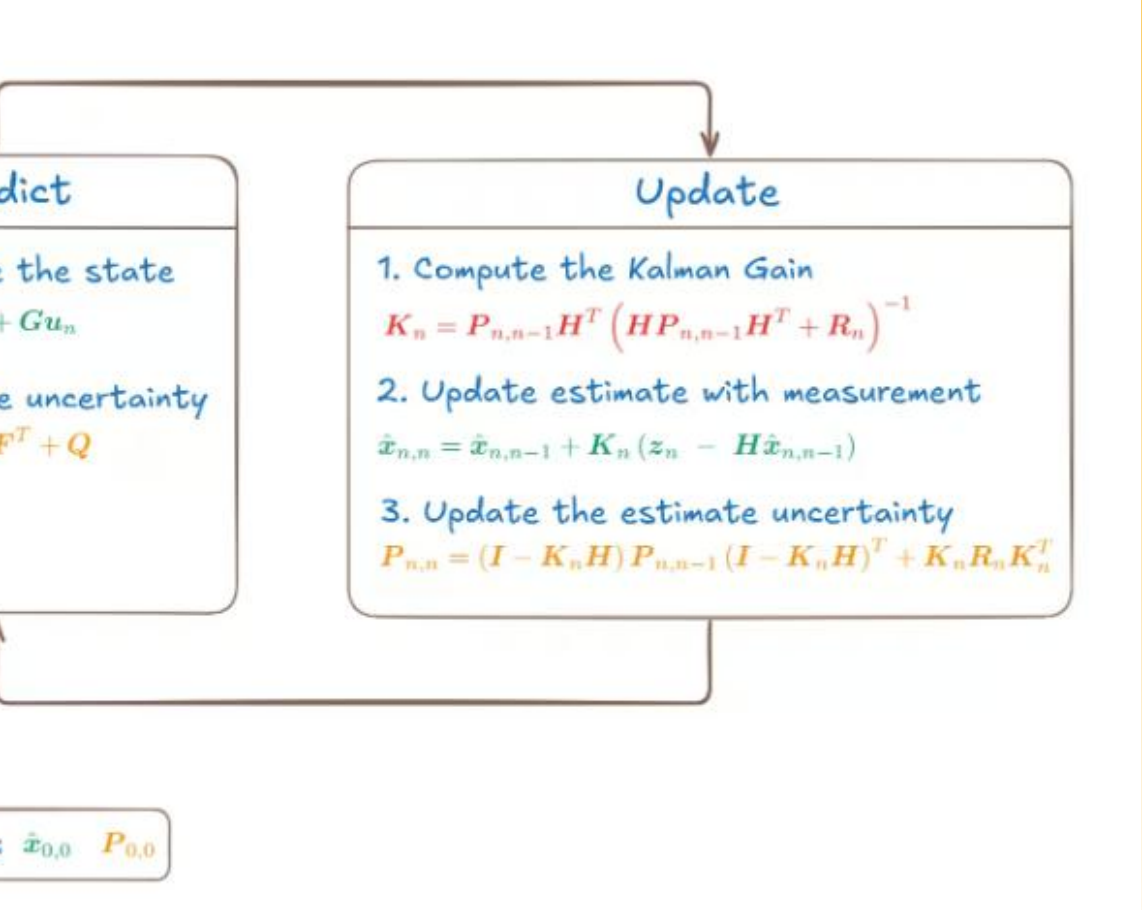


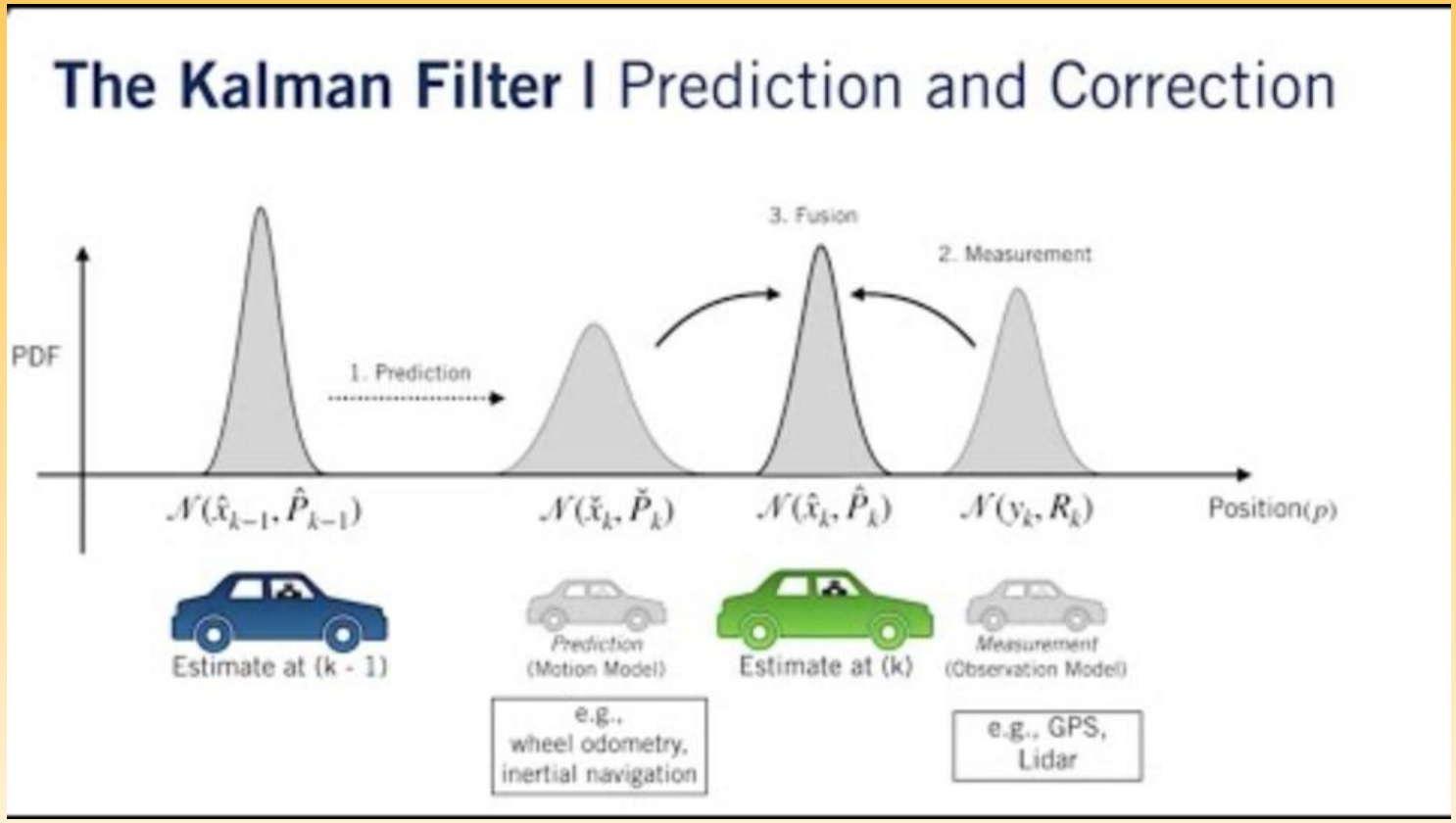
Figure 9. An AHRS feedback controller estimates the body orientation by fusing high-bandwidth gyro angular rate measurements with low-bandwidth attitude references. The yaw reference comes from a magnetometer or a GPS course-based estimate. Pitch and roll references are acquired from an estimate of the gravity vector via centripetally corrected accelerometer measurements. Essentially, this feedback controller uses a compensator to estimate the gyro biases by regulating the error between the estimated orientation and the orientation expressed by the low-bandwidth attitude references.

Sensor Fusion and Kalman Filters

- Sensor fusion techniques allow:
  - Common state based upon multiple sensor inputs
  - Can correct or mitigate measurement error
  - Can utilize the best sensor data



Kalman Filter



Kalman Filter Explained Through Examples

- Tutorial walk through.

Linear Kalman Filter

- Assume, we have the following data available:
  - GPS ($x_{E,GPS}, y_{E,GPS}, z_{E,GPS}$)
  - Air Data System (V_{ADS})
  - Altimeter ($z_{E,alt}$)
  - Compass (ψ_{Comp})
- Measurement Vector: $z_n = \begin{bmatrix} x_{E,GPS} & y_{E,GPS} & z_{E,GPS} & V_{ADS} & z_{E,Alt} & \psi_{Comp} \end{bmatrix}^T$
- State Vector: $\hat{\boldsymbol{x}} = \begin{bmatrix} x_E & y_E & z_E & V & \psi \end{bmatrix}^T$
- Note: as the book points out, this is a simplified example and more data would be needed for UAS navigation

Linear Kalman Filter (2)

- Several Matrices are utilized for the algorithm.
- State Transition Matrix
  - Relates how state will change after each change of a time increment, dT
  - E.g.
    - F =

$$\begin{bmatrix} 1 & 0 & 0 & -V \cdot s\psi \cdot \mathrm{d}t & 0 \\ 0 & 1 & 0 & V \cdot c\psi \cdot \mathrm{d}t & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- H Matrix
  - Relates measurement vector to the preceding state vector
  - E.g.

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Recall:

$$z_n = \begin{bmatrix} x_{\mathrm{E,GPS}} & y_{\mathrm{E,GPS}} & z_{\mathrm{E,GPS}} & V_{\mathrm{ADS}} & z_{\mathrm{E,Alt}} & \psi_{\mathrm{Comp}} \end{bmatrix}^T$$

$$\hat{\boldsymbol{x}} = \begin{bmatrix} x_E & y_E & z_E & V & \psi \end{bmatrix}^T$$

Linear Kalman Filter (3)

- Three covariance matrices are used:
  - Covariance Matrix, P
    - Quantifies state vector errors (calculated in Kalman filter algorithm)
  - Measurement noise covariance matrix, R
    - Expected measurement errors
    - Also known as Sequence Covariance Matrix
    - Sometimes, initially sensor errors populated with the standard deviations

$$R = \begin{bmatrix} \sigma^2_{\mathrm{GPS,xy}} & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma^2_{\mathrm{GPS,xy}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma^2_{\mathrm{GPS,z}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2_{\mathrm{ADS,V}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2_{\mathrm{Alt,z}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma^2_{\mathrm{Comp},\psi} \end{bmatrix}$$

- Also, covariance matrix of random system disturbances, Q
  - Bounds the UAS expected errors

$$Q = \begin{bmatrix} \sigma^2_{x,E} & 0 & 0 & 0 & 0 \\ 0 & \sigma^2_{y,E} & 0 & 0 & 0 \\ 0 & 0 & \sigma^2_{z,E} & 0 & 0 \\ 0 & 0 & 0 & \sigma^2_{V} & 0 \\ 0 & 0 & 0 & 0 & \sigma^2_{\psi} \end{bmatrix}$$

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Linear Kalman Filter (4)

- Now, let's walk through the algorithm
  - Two parts:
    - **Update Procedure** – updates the state matrix based upon sensor updates
    - **Extrapolation Procedure** – extrapolates the state matrix based upon time interval, dT

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Linear Kalman Filter (5)

- **Extrapolation Procedure:**
  - Allows extrapolation of the state between time steps even if no new measurements made. i.e. state updates at a higher rate than measurements through estimation
  - **Extrapolation Step 1:** Update state using state transition matrix

$$\hat{\boldsymbol{x}}_{n+1,n} = \boldsymbol{F}\hat{\boldsymbol{x}}_{n,n}$$

  - **Extrapolation Step 2:** Update the covariance matrix using the state transformation matrix and the random system disturbance covariance matrix.

$$\boldsymbol{P}_{n+1,n} = \boldsymbol{F}\boldsymbol{P}_{n,n}\boldsymbol{F}^T + \boldsymbol{Q}$$

Linear Kalman Filter (6)

- **Update Procedure**
  - **Update Step 1:** Compute Kalman Gain

$$K_n = P_{n,n-1} H^T \left(H P_{n,n-1} H^T + R_n \right)^{-1}$$

    - Where, dash superscript implies previous matrix prior to update

  - **Update Step 2:** Update State Given New Measurements and Preceding Estimate

$$\hat{x}_{1,1} = \hat{x}_{1,0} + K_1 (z_1 - H \hat{x}_{1,0})$$

  - **Update Step 3:** Update the Covariance Matrix, P

$$P_{n,n} = (I - K_n H) P_{n,n-1}$$

Kalman Filter Example

- Book example walks through set up of Kalman Filter for a rocket sled.

Sensor	Measurement	Error	Update Rate, Hz
Accelerometer	a_x	$0.1\ m/s^2$	20
GPS	x_E	12 m	1
Velocity encoder	u	0.001 m/s	10

- Need to define:
  - State Vector
  - Measurement Vector
  - State Transition Matrix
  - Q matrix
  - H matrix
  - R matrix

Kalman Filter Example (2)

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Accelerometer	a_x	0.1 m/s^2	20
GPS	x_E	12 m	1
Velocity encoder	u	0.001 m/s	10

- State and Measurement Vector:

$$\hat{\mathbf{x}} = \begin{bmatrix} x_E \\ u \\ a_x \end{bmatrix}$$

$$z_n = \begin{bmatrix} x_{\text{GPS}} \\ u_{\text{Enc}} \\ a_{\text{x,Accel}} \end{bmatrix}$$

Kalman Filter Example (2)

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Accelerometer	a_x	0.1 m/s^2	20
GPS	x_E	12 m	1
Velocity encoder	u	0.001 m/s	10

- The state transition matrix is based upon the kinematics of the rocket sled. We have equations for velocity and position measurement

$$V = V_0 + a_x \cdot \mathrm{d}t$$

$$x = x_0 + V_0 \cdot \mathrm{d}t + \frac{1}{2} \cdot a_x \cdot \mathrm{d}t^2$$

$$F = \begin{bmatrix} 1 & \mathrm{d}t & 1/2 \cdot \mathrm{d}t^2 \\ 0 & 1 & \mathrm{d}t \\ 0 & 0 & 1 \end{bmatrix}$$

Kalman Filter Example (2)

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GPS	x_E	12 m	1
Velocity encoder	u	0.001 m/s	10

- The H, R, and Q matrices are as follows:

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} \sigma^2_{\text{GPS},x} & 0 & 0 \\ 0 & \sigma^2_{\text{Enc}} & 0 \\ 0 & 0 & \sigma^2_{\text{Accel}} \end{bmatrix}$$

$$Q = \begin{bmatrix} \sigma^2_{x,E} & 0 & 0 \\ 0 & \sigma^2_{u} & 0 \\ 0 & 0 & \sigma^2_{a,x} \end{bmatrix}$$

Questions?

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